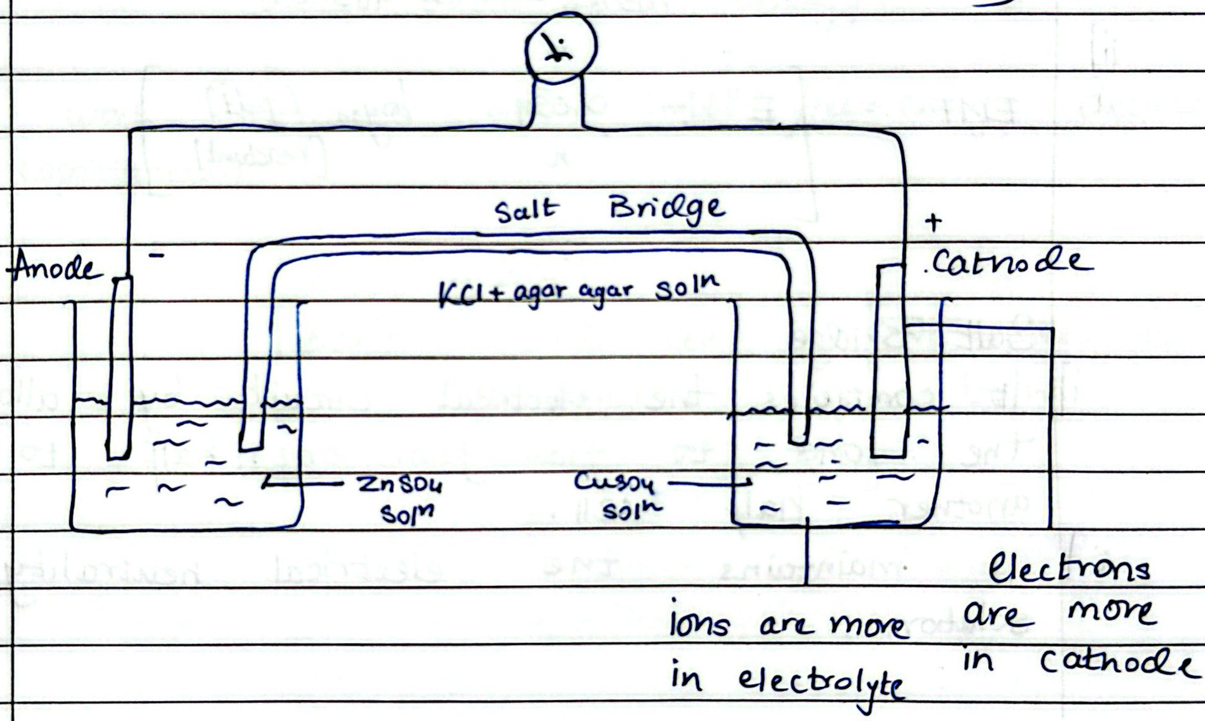
Chapter ~ 2

Electrochemistry



Standard Electrode Potential (E°)

EMF = $E_C - E_A$ or $E^\circ = E_C - E_A$.

Daniel Galvanaic Cell $\Longrightarrow$ Zn & Cu

$\boxed{SRP = - SOP}$

Cell representation : $Zn / Zn^{+2} \, // \, Cu^{2+} / Cu$

Phase Boundary

All Electrode Potential values are Standard

Standard form ⇒ 1 molal — conc.
1 atm — pressure
298 K — temp.

Formulae for EMF.

i) $$EMF = \left[E^\circ - \frac{2.303 R T}{n F} \log_{10} \frac{[pdt]}{[reactant]} \right]$$

8.314 (R); 298 K (T); moles (n); 96500 (F)

ii) $$EMF = \left[E^\circ - \frac{0.059}{n} \log_{10} \frac{[pdt]}{[reactant]} \right]$$

Salt Bridge

i) It completes the electrical circuit by allowing the ions to flow from one half to another half cell.

ii) It maintains the electrical neutrality of solution.

$w = q \times V$ — potential

└ Charge

1 mole of e^- ⟶ 1F
n mole of e^- ⟶ nF

∴ work done ⟹ $w = nFE^\circ$

(spontaneous Reaction) — by the system.
uses Gibbs Free energy.

work increases ⟹ Gibbs free energy decreases
(spontaneous)

$$W = -\Delta G$$

$$\boxed{\Delta G = -nFE^\circ_{cell}}$$

At Equilibrium

$$\boxed{\Delta G = -2.303\ RT\ \log K_c}$$

-ve value in both

⇒ Electrolytic Conductance

1) Resistance (R) ⇒ $V = IR$.

2) Conductance (C) ⇒ $C = \frac{1}{R}$.

3) Specific Resistance (ρ) ⇒ $\frac{RA}{l}$ ⇒ distance b/w electrodes

$\frac{l}{A}$ = Cell Constant or $\boxed{\frac{l}{A} = \kappa \times R}$

4) Conductivity (κ) / Specific Conductance

Kappa

i) ⇒ $\frac{1}{\rho} = \frac{1}{\frac{RA}{l}} = \frac{l}{RA}$

ii) ⇒ conductance × Cell constant

⇒ $\frac{1}{R} \times \frac{l}{A}$.

5) Molar Conductivity (Λm)

i) ⇒ $\frac{\kappa \times 1000}{\text{molarity}}$ or

ii) ⇒ $\kappa \times$ Volume

6) Equivalent Conductivity (Λeq)

⇒ $\frac{\kappa \times 1000}{\text{normality}}$

Normality = $\frac{\text{given mass}}{\text{equivalent mass}}$ eq.mass = $\frac{\text{mol. mass}}{\text{valency}}$.

Note: Degree of dissociation (α) = $\frac{\Lambda m}{\Lambda m^{\circ}}$ — given / — calculated

=> Kohlraush's Law

$$\Lambda_m^{\infty} = v_+ \lambda_+^{\infty} + v_- \lambda_-^{\infty}$$

=> Faraday's Law

=> $Q = \underset{\text{Current}}{I}T \longrightarrow$ time (in secs)

=> $m = \underbrace{Z}_{\text{Electrochemical Equivalent}}Q$

=> $w = ZIT \,/\, F \rightarrow 96500$

=> $m = \frac{EIT}{nF}$

⟹ Numericals Solutions

* Constants

i) $R = 8.314$ J/mol/K
$F = 96500$
$T = 298$ K

Formulae :- $\left[E^\circ - \frac{2.303RT}{nF} \log_{10} \frac{[pdt]}{[reactant]} \right]$

$= \left[E^\circ - \frac{0.059}{n} \log_{10} \frac{[pdt]}{[react]} \right]$

⟹ EMF Numericals

1) E° Cell from standard Electrode Potentials

Q.1 Calculate · Calculate E° for the cell
$Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$

$E\ (Cu^{2+}/Cu) = 0.34V.$
$E\ (Zn^{2+}/Zn) = -0.76V$

[Idea] Always the positive value or greater value is cathode
Thus,

$E^\circ = E_C - E_A$
$E^\circ = 0.34 - (-0.76)$
$\boxed{E^\circ = 1.10V.}$

2.] Type 2 : Type 2 Nernst Equation. under non-standard conditions

Question $Zn | Zn^{2+} (0.1M) \| Cu^{2+} (1.0M) | Cu$

$E^0 = 1.10V$

Calculate : EMF.

Answer: Write Equation.

$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu.$

$n = 2.$ (as both half reactions have 2).

$Q = \frac{pdt}{reactant}$

$Q = \frac{0.1}{1} = 0.1$

$EMF = 1.10 - \frac{0.0591}{2} \times \log(0.1)$

$\boxed{= 1.1296V.}$

3] For the cell $Ni / Ni^{2+} (1.0M) \| Ag^+ (x M) | Ag$

$E^0 cell = 1.05V$

$E^0 (Ag^+/Ag) = 0.80V$ $E^0 (Ni^{2+}/Ni) = -0.25V$

$EMF = 1.00V.$

$n = 2.$

$Q = \frac{1}{[Ag^+]^2}$

$E_{cell} = E^0 - \left(\frac{0.0591}{n}\right) \log \left(\frac{1}{Ag^{+2}}\right)$

$$1 = 1.09 - \left(\frac{0.0591}{2}\right) \log \left(\frac{1}{Ag^{+2}}\right)$$

$$0 = 1.692 \log [Ag^{+}]$$

$$= 10^{\wedge} - 0.846$$

$$\boxed{= 0.143\ M}$$

4.

⇒ Type 4 : Concentration Cells

Question) Calculate E° cell for the cell $Cu \,|\, Cu^{2+}\,(0.01M) \,||\, Cu^{2+}\,(1.0M) \,|\, Cu\,(s)$ at 298 K.

Note: As this is a concentration cell $E^{\circ}cell = 0$.
(The electrode which is more
dilute = 0.01M anode.
concentrate = 1.0M cathode)

$$Q = \frac{0.01}{1} = \boxed{0.01}$$

$$EMF = 0 - \left(\frac{0.0591}{2}\right) \log (0.01)$$

$$= -(0.02955) \times (-2)$$

$$\boxed{= 0.0591\ V}$$

5) ~~Type~~ 5 .

⟹ Calculate the electrode potential of a Zn electrode dipped in 0.01 M $ZnSO_4$ soln. Given (E° $(Zn^{2+}/Zn) = -0.78V$.

$n=2$

Idea : For a single electrode the formula changes to reduction

$$\boxed{EMF = E^\circ + \left(\frac{0.0591}{n}\right) \times \log(\text{ion conc.})}$$

6) Type 6 :

⟹ The standard cell potential for a reaction is $E^\circ cell = 0.295 V$ with $n=2$, Calculate equilibrium constant K_c at 298k.

Idea : At Equilibriam $E^\circ cell = 0$. and $Q = K_c$.

$$0 = E^\circ - \left(\frac{0.0591}{n}\right) \log K_c.$$

$$\log K_c = \frac{2 \times 0.295}{0.0591}$$

$$\log K_c = 9.983$$

$$\boxed{K_c = 9.62 \times 10^9}$$

⟹ GIBBS FREE ENERGY NUMERICALS.

1) $\Delta G = \Delta H - T\Delta S$ (spontaneity and transition temperature)

Question) $\Delta H = -92.0$ kJ/mol $= -92000$ J/mol
$\Delta S = -198.0$ J/K mol
$T = 298$ K

$$\Delta G = \Delta H - T\Delta S$$
$$= -92000 - (298 \times (-198))$$
$$= -32996 \text{ J/mol}$$
$$\boxed{= -3.2996 \text{ kJ/mol}}$$

Note) To find the temp at which it will become non-spontaneous.

Idea: For that $\Delta G = 0$.

$$0 = -92000 - T \times (-198)$$
$$\boxed{T = 464.6 \text{ K}}$$

2) Type 2: Calculate ΔG when K_c is given

$K_c = 10$
$T = 298$K

$$\Delta G = -2.303RT \log K_c$$
$$= -2.303 \times 8.314 \times 298 \times \log(10)$$
$$= -5705.8 \text{ J/mol}$$
$$\boxed{= -5.705 \text{ kJ/mol}}$$

3) ΔG and EMF (the bridge formula)

Question) $E^\circ = 1.10V$

$n = 2.$

$\Delta G = ?$

$$\Delta G = -nFE^\circ$$

$$\Delta G = -2 \times 96500 \times 1.10$$

$$\boxed{\Delta G = -212{,}300 \ J/mol}$$

$$= -212.3 \ kJ/mol.$$

4) Type 4 — ΔG (non-standard) using E cell

$E^\circ = 1.1296V$

$n = 2$

$\Delta G = ?$

$$\Delta G = -nFE^\circ$$

$$\Delta G = -2 \times 96500 \times 1.1296 = -217900$$

$$\boxed{\delta G = -217.9 \ kJ/mol}$$